Hybrid Identity Lab — Build Log

Oct 1, 2026 · @Pob

Overview

An on-prem Active Directory lab on Proxmox, built to show hands-on identity skills (AD, Group Policy, and later Entra ID and AWS) for DevOps and cloud roles. It is aimed at employers, not at the Obnix app; Obnix's domain is only used as a realistic UPN suffix.

Status (Oct 14): Phases 1 to 3 are complete. The five lab users sync from AD to Entra ID, Azure access follows the synced groups (just-in-time Contributor for IT through PIM, standing Reader for Finance), and Conditional Access enforces MFA and blocks legacy authentication. Phase 4 (app SSO) is next.

End goal: hybrid identity. On-prem AD syncs to Entra ID through Entra Connect, and Entra ID provides SSO to Azure, AWS IAM Identity Center, Grafana and Argo CD. Disabling a user once in AD removes access everywhere.

Project phases

The lab is built in five phases; each one ends with a test that proves it works and can be shown on its own.

Phase	Focus	Done when	Status
1. On-prem AD foundation	Domain, DNS, OU, groups, users, lockout policy, targeted GPO, domain-joined client	Akua (Finance) blocked from Control Panel, Ama (IT) allowed	Done
2. Hybrid identity	Entra ID tenant, verify `lab.obnixapp.com`, Entra Cloud Sync with password hash sync	A user disabled in AD can no longer sign in to Entra	Done
3. Cloud access control	Azure RBAC through synced groups with PIM, Conditional Access and MFA	IT-Staff can manage the resource; Finance-Staff can't	Done

Phase	Focus	Done when	Status
4. Multi-cloud and app SSO	Entra to AWS IAM Identity Center (SAML + SCIM); SSO into Grafana, Argo CD, Proxmox	One login reaches Azure, AWS and the homelab apps	Not started
5. Monitoring and hardening	Prometheus/Grafana alerts for DC health and sign-in failures; LDAPS and LDAP signing/channel binding; gMSA for remaining service accounts, LAPS, smart card logon	The identity layer is monitored, LDAP is encrypted and signed, and admin access is hardened	Not started

Phase 3 runs on a 30-day Entra ID P2 trial, licensed by group, which ends around Nov 3, 2026. Basic SAML/OIDC SSO in Phase 4 doesn't need P2.

Phase 1: On-prem AD foundation

Completed Oct 2, 2026. A domain controller (wind01) and a domain-joined Windows 11 client (CLIENT01), with users, groups, a lockout policy and a targeted GPO, all verified on the client.

Domain controller build (wind01)

wind01 (VM 301 on pve3) is a Windows Server 2022 Standard Evaluation DC, promoted to a new forest `homelab.internal` (NetBIOS `HOMELAB`, functional level Windows Server 2016) with AD DS and DNS. The DSRM password was set at a hidden prompt.

Setting	Value
CPU	2 cores, type x86-64-v2-AES
Memory	6 GB
Firmware / machine	SeaBIOS, i440fx
Disk	104 GB on local-lvm, IDE (ide0)
Network	Intel e1000 on vmbr0

It uses emulated hardware (IDE, e1000), so no VirtIO drivers were needed at install. Moving it to VirtIO disk and NIC is a follow-up.

```
Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools
$dsrm = Read-Host -AsSecureString "DSRM password"
Install-ADDSForest -DomainName "homelab.internal" -DomainNetbiosName
"HOMELAB" `
    -ForestMode "WinThreshold" -DomainMode "WinThreshold" `
    -InstallDns -SafeModeAdministratorPassword $dsrm -Force
```

Environment inventory

Item	Value
Domain	`homelab.internal` (NetBIOS `HOMELAB`)
Domain controller	`WIN-HJ2P7RVTJ88`, VM 301 `wind01` on pve3, IP 192.168.40.216 (reserved on the router). Windows Server 2022 Standard Evaluation (rearmed Oct 2). Roles: AD DS, DNS, File and Storage. Time from pool.ntp.org
Client	`CLIENT01`, VM 302 on pve, Windows 11 Enterprise LTSC 2024 Evaluation (90 days), IP 192.168.40.103 (reserved on the router), joined to the domain
UPN suffix	`lab.obnixapp.com`

Node capacity (Oct 1)

Node	RAM	Used	Notes
pve	15.5 GB	44%	Most headroom; Proxmox 9.2.3. Hosts CLIENT01
pve2	7.6 GB	83%	k8s-node1, wordpress, adguard, local-app-node; Proxmox 9.1.1
pve3	7.6 GB	97%, 3.75 GB swap	Overloaded: DC, k8s-node2; Proxmox 9.1.1

The DC still has its default hostname; renaming a DC needs extra steps and is deferred.

Directory structure

One OU, three department groups and five test users, all created with PowerShell on Oct 1. Users log in with email-style names and must change the temporary password at first logon.

User	Login (UPN)	Group
Ama Mensah	ama.mensah@lab.obnixapp.com	IT-Staff
Kwame Boateng	kwame.boateng@lab.obnixapp.com	IT-Staff
Kofi Asante	kofi.asante@lab.obnixapp.com	Finance-Staff
Akua Darko	akua.darko@lab.obnixapp.com	Finance-Staff
Esi Owusu	esi.owusu@lab.obnixapp.com	HR-Staff

OU path: `OU=LabUsers,DC=homelab,DC=internal`. Groups are Global scope, inside the same OU.

```
Get-ADForest | Set-ADForest -UPNSuffixes @{Add="lab.obnixapp.com"}
New-ADOrganizationalUnit -Name "LabUsers" -Path "DC=homelab,DC=internal"
"IT-Staff","Finance-Staff","HR-Staff" | ForEach-Object {
  New-ADGroup -Name $_ -GroupScope Global -Path
"OU=LabUsers,DC=homelab,DC=internal" }

# Users from C:\lab-users.csv (First,Last,Dept); temp password entered with
Read-Host
Import-Csv C:\lab-users.csv | ForEach-Object {
  $sam = "$($_.First).$($_.Last)".ToLower()
  New-ADUser -Name "$($_.First) $($_.Last)" -GivenName $_.First -Surname
$_.Last `
    -SamAccountName $sam -UserPrincipalName "$sam@lab.obnixapp.com" `
    -EmailAddress "$sam@lab.obnixapp.com" -Department $_.Dept `
    -Path "OU=LabUsers,DC=homelab,DC=internal" `
    -AccountPassword $tmp -ChangePasswordAtLogon $true -Enabled $true
  Add-ADGroupMember -Identity "$($_.Dept)-Staff" -Members $sam
}
```

The email attributes are login names only; no mailboxes exist until users sync to Microsoft 365.

Password and lockout policy

The default domain policy was left as installed, except lockout: accounts now lock for 15 minutes after 5 wrong passwords (previously never).

Setting	Value
Complexity	On: 3 of 4 character types, no parts of the account name
Minimum length	7
Maximum age	42 days
Minimum age	1 day (an admin `-Reset` bypasses it)
History	24 passwords
Lockout threshold	5 attempts (was 0, changed Oct 1)
Lockout duration / window	15 minutes

```
Set-ADDefaultDomainPasswordPolicy -Identity homelab.internal -
LockoutThreshold 5 `
  -LockoutDuration 00:15:00 -LockoutObservationWindow 00:15:00
```

Group Policy: Lab-Restrict-ControlPanel

A user GPO that hides Control Panel for Finance and HR staff but not IT. It is linked to the LabUsers OU and narrowed with security filtering. Created and tested Oct 1 on CLIENT01: Akua Darko (Finance) gets "This operation has been cancelled due to restrictions"; Ama Mensah (IT) opens Control Panel normally. First domain logons were done on the Proxmox console, since users must change their temporary password and RDP blocks that by default.

Trustee	Permission
Finance-Staff	GpoApply
HR-Staff	GpoApply
Authenticated Users	GpoRead (read only, so it does not apply to everyone)

Trustee	Permission
Domain Admins, Enterprise Admins, SYSTEM	GpoEditDeleteModifySecurity
Enterprise Domain Controllers	GpoRead

Setting: `HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer` → `NoControlPanel` = 1 (DWord).

```
New-GPO -Name "Lab-Restrict-ControlPanel" | New-GPLink -Target
"OU=LabUsers,DC=homelab,DC=internal"
Set-GPRegistryValue -Name "Lab-Restrict-ControlPanel" `
  -Key "HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer" `
  -ValueName "NoControlPanel" -Type DWord -Value 1
Set-GPPermission -Name "Lab-Restrict-ControlPanel" -TargetName "Authenticated
Users" `
  -TargetType Group -PermissionLevel GpoRead -Replace
"Finance-Staff","HR-Staff" | ForEach-Object {
  Set-GPPermission -Name "Lab-Restrict-ControlPanel" -TargetName $_ -
TargetType Group -PermissionLevel GpoApply }
```

Authenticated Users keeps Read because computers must be able to read every GPO (since MS16-072); only Apply is narrowed.

Client VM build: CLIENT01

The lab started with one Windows VM: **wind01** (VM 301 on pve3), the domain controller that runs Active Directory and DNS for `homelab.internal`. A DC can't serve as a test machine, because regular domain users are not allowed to log on to a domain controller. Testing user logons and user Group Policy needs a separate, domain-joined workstation, so CLIENT01 was built to play the role of an employee's PC.

CLIENT01 went on pve because pve3 (with the DC) was at 97% RAM and pve2 at 83%. The two VMs sit on the same LAN (192.168.40.0/24), so placement doesn't affect how they talk.

How CLIENT01 depends on wind01

Service on wind01	How CLIENT01 uses it
DNS	CLIENT01's only DNS server is 192.168.40.216; it finds the domain through DNS records the DC publishes
Domain join	CLIENT01 has a computer account in AD and a trust relationship with the domain
Authentication (Kerberos)	Every domain logon (Akua, Ama) is checked by the DC
Group Policy (SYSVOL)	At logon, CLIENT01 downloads and applies the GPOs that target the user
Password policy	Complexity, history and lockout rules come from the domain, not the client

ISOs (local storage on pve)

- Windows 11 Enterprise LTSC 2024 Evaluation, x64, 4.8 GB, from Microsoft's Evaluation Center. LTSC was chosen because it is leaner (no Store apps), which suits the tight RAM.
- `virtio-win-0.1.285.iso` (already on pve).

VM settings (VM 302)

Setting	Value
OS type	Windows 11/2022/2025, VirtIO driver ISO as second CD
Machine / BIOS	q35, OVMF (UEFI), EFI disk + TPM 2.0 on local-lvm
Disk	64 GB SCSI, VirtIO SCSI single, discard on
CPU	2 cores, type host (all nodes share the i5-6500T)
Memory	4096 MB, ballooning on (min 2048)
Network	VirtIO on vmbr0
Guest agent	Enabled

Step-by-step build

1. On the laptop, download the Windows 11 Enterprise LTSC 2024 Evaluation ISO from Microsoft's Evaluation Center (free registration form, no product key).
2. In Proxmox: pve → local (pve) → **ISO Images** → Upload the Windows ISO. Confirm `virtio-win-0.1.285.iso` is in the same list.
3. **Create VM** with the settings in the table above, attaching the VirtIO ISO as the second CD on the OS tab.
4. Start the VM, open the Console, and press a key at "Press any key to boot from CD."
5. Setup: language → Install Windows 11 → image **Windows 11 Enterprise LTSC Evaluation** (not the N edition) → accept the license.
6. At "Select location to install", click **Load driver** → VirtIO CD → `vioscsi\w11\amd64` → Red Hat VirtIO SCSI pass-through controller. The 64 GB disk appears.
7. Load driver again → `NetKVM\w11\amd64` → Red Hat VirtIO Ethernet Adapter, so the VM has network after install.
8. Select Disk 0 → Next. Windows installs and reboots on its own (about 15 minutes).
9. Out-of-box setup: Sign-in options → **Domain join instead** → local account `labadmin`, password, three security questions; all privacy toggles off; **Update later**.
10. At the desktop, `ipconfig` showed 192.168.40.103 from the router.
11. Settings → System → **Remote Desktop: On**. Connect from the laptop with `mstsc` as `.\labadmin`.

Install gotchas

- The "Press any key to boot from CD" prompt lasts about 5 seconds. Missing it falls through to PXE and HTTP boot; after those time out, boot the DVD from the UEFI Boot Manager.
- No disk appears until the storage driver is loaded: `vioscsi\w11\amd64`. The network driver (`NetKVM\w11\amd64`) was loaded at the same screen.
- ISOs must go in the ISO Images tab of local storage, not CT Templates.

Domain join

1. Local account `labadmin` created at setup (Sign-in options → Domain join instead). Windows updates deferred.
2. Remote Desktop enabled; RDP from the laptop as `.\labadmin` for copy and paste (the Proxmox console can't paste).

3. DNS pointed at the DC, then `Resolve-DnsName homelab.internal` confirmed 192.168.40.216.
4. `Add-Computer -NewName CLIENT01` joined the domain but the rename failed with "The directory service is busy" — a known quirk of join + rename in one step. Fixed with a restart, then `Rename-Computer -DomainCredential HOMELAB\Administrator -Restart`.
5. Verified: Name CLIENT01, Domain homelab.internal, PartOfDomain True.

The VirtIO guest-tools installer stalled at "Preparing to install"; the network driver was already loaded during setup, so it was deferred.

```
Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses
192.168.40.216
Resolve-DnsName homelab.internal
Add-Computer -DomainName homelab.internal -Restart
Rename-Computer -NewName CLIENT01 -DomainCredential HOMELAB\Administrator -
Restart
```

User logon, password change and GPO verification

Each test user was created with a temporary password and "change password at next logon", so the first logon forces a new password. RDP refuses that first logon by default, so it was done on the Proxmox console.

1. On the DC, set a fresh temporary password if the original is unknown:

   ```
   $tmp = Read-Host -AsSecureString "New temporary password"
   "akua.darko","ama.mensah" | ForEach-Object {
     Set-ADAccountPassword $_ -Reset -NewPassword $tmp
     Set-ADUser $_ -ChangePasswordAtLogon $true }
   ```

2. Proxmox console on CLIENT01 → sign out of labadmin → **Other user** → `akua.darko@lab.obnixapp.com` with the temporary password → set a new password when prompted. Windows builds her profile on first logon.
3. Right-click Start → Run → `control`. Result: "This operation has been cancelled due to restrictions in effect on this computer." The GPO applies to Finance.
4. Sign out → Other user → `ama.mensah@lab.obnixapp.com` → change password.
5. Run → `control`. Result: Control Panel opens normally. The GPO does not apply to IT.

User	Group	First logon	Control Panel
Akua Darko	Finance-Staff	Password changed	Blocked by GPO
Ama Mensah	IT-Staff	Password changed	Opens normally

Troubleshooting if a policy doesn't apply: run `gpupdate /force` as the user, sign out and back in, then check `gpresult /r` for the applied GPOs.

Phase 1 lessons learned

- **DNS comes first.** A client must use the DC as its DNS server; `Resolve-DnsName homelab.internal` before joining saves a failed join.
- **Join and rename in separate steps.** A one-step `Add-Computer -NewName` joined the domain but failed the rename with "The directory service is busy."
- **Security filtering narrows Apply, not Read.** Authenticated Users keeps Read so computers can process the GPO.
- **Test both sides of a policy.** One user it should hit (Akua, Finance) and one it should not (Ama, IT).
- **First domain logon needs a password change,** which RDP blocks by default; do it on the console.
- **Check capacity before placing VMs.** pve3 was at 97% RAM; the client went on pve, which had room.
- **Know the complexity rules.** AD rejects passwords containing 3+ character pieces of the account or display name.

Interview story (about 30 seconds): *"I built an on-prem Active Directory lab on Proxmox: a domain controller, an OU with department groups and users created in PowerShell, an account lockout policy, and a GPO targeted with security filtering. I joined a Windows 11 client and proved the policy: Finance users were blocked from Control Panel, IT users weren't. Next I'm syncing it to Entra ID for hybrid SSO across Azure, AWS and my monitoring tools."*

Phase 1 close-out

- [x] ~~Finish the CLIENT01 install with a local account~~
- [x] ~~Record the DC's IP (192.168.40.216)~~
- [x] ~~Join CLIENT01 to `homelab.internal`~~

- [x] ~~Test the GPO: Akua blocked, Ama allowed~~
- [x] ~~Readiness checks on the DC: `dcdiag /q`, group memberships, UPNs on `lab.obnixapp.com`, functional levels and OS version, time sync (`w32tm /query /status`)~~
- [x] ~~Proxmox snapshot of wind01 before installing Entra Connect (named pre-entra-connect, Oct 2, includes RAM; delete once Phase 2 is stable)~~
- [x] ~~Reserve .216 (DC) and .103 (CLIENT01) on the router~~
- [ ] Finish VirtIO guest tools (`D:\virtio-win-gt-x64.msi`)
- [ ] Grant RDP access to a lab group (domain users can't RDP by default)
- [x] ~~Rearm the DC evaluation before Phase 2: 55 days left on Oct 1 (expires about Nov 26), 6 rearms available. `slmgr /rearm`, reboot, confirm about 180 days with `slmgr /dlv`~~

Readiness results (Oct 1): `dcdiag /q` clean; IT 2, Finance 2, HR 1 members; all UPNs on `lab.obnixapp.com`; Windows2016Forest on Server 2022 Standard Evaluation. The evaluation had 55 days left (6 rearms available) and was rearmed on Oct 2 for a fresh 180 days, so it won't expire mid-Phase 2. Time sync failed the check: the DC ran on a free-running local clock. Fixed by pointing it at external NTP; it now syncs from pool.ntp.org at stratum 3.

```
w32tm /config /manualpeerlist:"time.windows.com,0x8 pool.ntp.org,0x8"
/syncfromflags:manual /reliable:yes /update
Restart-Service w32time
w32tm /resync /rediscover
```

Phase 2: Hybrid identity

Completed Oct 3, 2026. On-prem AD now syncs to a Microsoft Entra ID tenant with Entra Cloud Sync and password hash sync, scoped to the LabUsers OU.

Item	Value
Tenant	Homelab Identity (Entra ID Free), initial domain `philipobourbonsuoutlookcom.onmicrosoft.com`
Custom domain	`lab.obnixapp.com`, verified with a TXT record in Cloudflare

Item	Value
Cloud admin	`admin@…onmicrosoft.com`, cloud-only Global Administrator; the personal Microsoft account stays as backup owner
Sync tool	Entra Cloud Sync agent on wind01, running as gMSA `homelab.internal\provAgentgMSA`
Scope	`OU=LabUsers,DC=homelab,DC=internal`
Password hash sync	Enabled
Accidental deletion protection	Enabled, threshold 500

Why Cloud Sync rather than Entra Connect Sync

Cloud Sync is a lightweight agent configured from the cloud, and Microsoft's direction for new deployments. Connect Sync runs a full sync engine with a local database, which wind01 (6 GB RAM on a node already at 97%) couldn't comfortably host. The concepts are the same: source of authority stays on-prem, password hash sync, OU scoping.

Build steps

1. **Tenant.** Signed up for Azure with the existing Microsoft account. Creating a new workforce tenant now needs a paid licence, and the External tenant type is for customer apps, so the account's existing workforce directory was used and renamed to Homelab Identity. The Azure free trial was not started; its 30-day credit is saved for Phase 3.
2. **Cleanup.** Deleted three leftover test users and turned off elevated "Access management for Azure resources" once it was no longer needed.
3. **Cloud admin.** Used the existing cloud-only `admin@` account (Global Administrator) for all lab work.
4. **Custom domain.** Added `lab.obnixapp.com` in Entra and created the `MS=ms…` TXT record in Cloudflare with name `lab`. Verified within minutes. The record proves ownership only; it doesn't touch the live app.
5. **Agent install on wind01.** Downloaded the provisioning agent from Cloud sync → Agents and ran the wizard: connect Entra (as `admin@`), create gMSA (with `HOMELAB\Administrator`), connect AD (`homelab.internal`).

6. **Configuration.** New configuration → AD to Microsoft Entra ID sync → `homelab.internal`, password hash sync on, scope set to the LabUsers OU.
7. **Provision on demand** for Akua: import, scope, match and create all succeeded before enabling the configuration.
8. **Enabled** the configuration; all five lab users appeared in Entra with on-premises sync enabled.

Problems worth noting

- **Microsoft account signup was blocked twice** ("unusual activity") from a private window. Used the existing Microsoft account instead; the tenant, not the account, is what gives separation.
- **`state_mismatch` sign-in error** after tenant switching. Fixed by signing out, closing all portal tabs and opening the tenant directly: `portal.azure.com/<tenant>.onmicrosoft.com`.
- **IE Enhanced Security Configuration** on Windows Server blocked the agent's sign-in window. Turned it off for Administrators during setup, then back on.
- **The wizard skipped the service account and AD pages** after the failed first run, so the summary showed only the Entra login. A clean reinstall of the agent fixed it; the summary then listed the gMSA and the domain.
- **Scope silently reverted to All objects**, so the first sync also pulled in `sql_service`. Setting the scope back to the LabUsers OU removed it from Entra on the next cycle; the AD account was untouched. Lesson: scoping decides exactly what leaves the building, so check it before enabling.

Verification: the Phase 2 test

Step	Result
Akua signs in to myapps.microsoft.com with her AD password	Success (password hash sync)
`Disable-ADAccount akua.darko` on wind01	Enabled = False in AD
Provision on demand for Akua	Successfully updated object in Entra ID
Akua signs in again	Blocked: "Your account has been locked. Contact your support person"

Step	Result
`Enable-ADAccount akua.darko`	"Enable account" synced at 1:51 AM; Akua signed in again with Authenticator MFA

One change in AD removed cloud access. That is the point of hybrid identity.

Phase 2 close-out

- [x] ~~Tenant renamed and cleaned up~~
- [x] ~~`lab.obnixapp.com` verified~~
- [x] ~~Cloud Sync agent installed and active, running as a gMSA~~
- [x] ~~Configuration scoped to LabUsers, password hash sync on~~
- [x] ~~Disable-in-AD test passed~~
- [x] ~~Re-enable Akua and confirm she can sign in again~~
- [ ] IE ESC back on for Administrators on wind01
- [ ] Set the DC time zone to Central (`tzutil /s "Central Standard Time"`)
- [ ] Delete the `pre-entra-connect` snapshot once sync has run cleanly for a few days

Phase 3: Cloud access control

Completed Oct 11, 2026. Azure access is controlled entirely by the AD-synced groups: IT gets just-in-time Contributor through PIM, Finance gets standing Reader, and Conditional Access enforces MFA for everyone except a break-glass account.

Step	Task	Status
1	Confirm AD groups synced to Entra (Security, Source: Windows Server AD)	Done
2	Azure subscription `lab-subscription` (pay-as-you-go) + $5 budgets	Done
3	Entra ID P2 trial, licensed by group in the M365 admin center	Done
4	RBAC on `rg-identity-lab`: IT-Staff Contributor (PIM eligible), Finance-Staff Reader (active)	Done, tested

Step	Task	Status
5	Conditional Access: break-glass, require MFA, block legacy auth, risk-based MFA	Done, tested

Problems worked through

- **No free credit.** The account's Azure free trial was already used in 2025, so billing was upgraded to pay-as-you-go. Expected cost is about $0: resource groups, role assignments and tags are free.
- **Subscription transfer blocked** (`blockSubscriptionsIntoTenant`). The cloud admin was temporarily elevated to accept ownership, then elevation was turned off instead of loosening the tenant policy.
- **PIM made everything eligible.** With P2 on, the portal created both role assignments as Eligible. Finance Reader was re-added as Active; write access stays just-in-time.
- **Identity Protection blocked Ama's first sign-in** (new user, new device, no MFA). The admin dismissed the risk, Ama registered Authenticator, and CA03 now lets users self-remediate with MFA.
- **Break-glass role was PIM-eligible.** It was changed to Active, permanent: an emergency account can't depend on PIM activation.

RBAC tests

Test	User	Result
Finance cannot change	Akua (Reader)	Sees the resource group; saving a tag fails with `AuthorizationFailed`
IT must activate first	Ama (eligible Contributor)	Activates Contributor in PIM for 1 hour; tag saved. The activity log shows MS-PIM creating the temporary assignment

Conditional Access

Policy	Scope	Control
CA01 Require MFA	All users except break-glass, all resources	Grant with MFA

Policy	Scope	Control
CA02 Block legacy auth	Same; client apps: Exchange ActiveSync + other clients	Block
CA03 Risky sign-ins	Same; sign-in risk medium and high	Grant with MFA

Security defaults were turned off. The break-glass account (cloud-only, Global Administrator active and permanent, long random password, its own MFA) was created before any policy.

Incident: locked out by CA02

On Oct 11 the admin account got error 53003 (blocked by Conditional Access) on the Azure portal. Break-glass still signed in. CA02 had saved with **0 conditions** and Block access, so it blocked every sign-in, not just legacy clients. It was turned off, the client-apps condition was fixed, the policy was verified in report-only mode with What If, and then turned back on. Admin sign-in recovered after about 25 minutes of propagation delay.

Verification	Result
What If: admin, Azure portal, browser	CA01 applies; CA02 and CA03 don't
What If: admin, Exchange ActiveSync	CA02 blocks
Live	Admin signs in with MFA with all three policies on

Full write-up and screenshots: `docs/phase-3-cloud-access-control.md` in the repo.

Phase 4: Multi-cloud and app SSO

- [ ] Entra as the identity source for AWS IAM Identity Center (SAML + SCIM)
- [ ] Entra SSO (OIDC) into Grafana, Argo CD and Proxmox

Phase 5: Monitoring and hardening

- [ ] windows_exporter on the DC; Prometheus alerts for DC down and failed-logon spikes, routed to PagerDuty and Discord

- [ ] gMSA for remaining service accounts (the Cloud Sync agent already runs as provAgentgMSA), LAPS for local admin passwords
- [ ] Smart card / YubiKey logon with AD Certificate Services; LDAPS and LDAP signing / channel binding on the DC

Homelab housekeeping

- [ ] Rebalance nodes: migrate a VM off pve3 to pve
- [ ] Update pve2 and pve3 to Proxmox 9.2.x to match pve
- [ ] Add RAM to one node (16–32 GB)
- [x] ~~Portfolio repo: pushed Oct 2 to github.com/pobournex/hybrid-identity-lab (private) with README, Phase 1 docs, scripts and PDF; add screenshots~~